

12. TIC TAC TOE GAME

Aim:

To design and implement a Tic Tac Toe game in Python using logical decision - making, board validation, and win/draw checking.

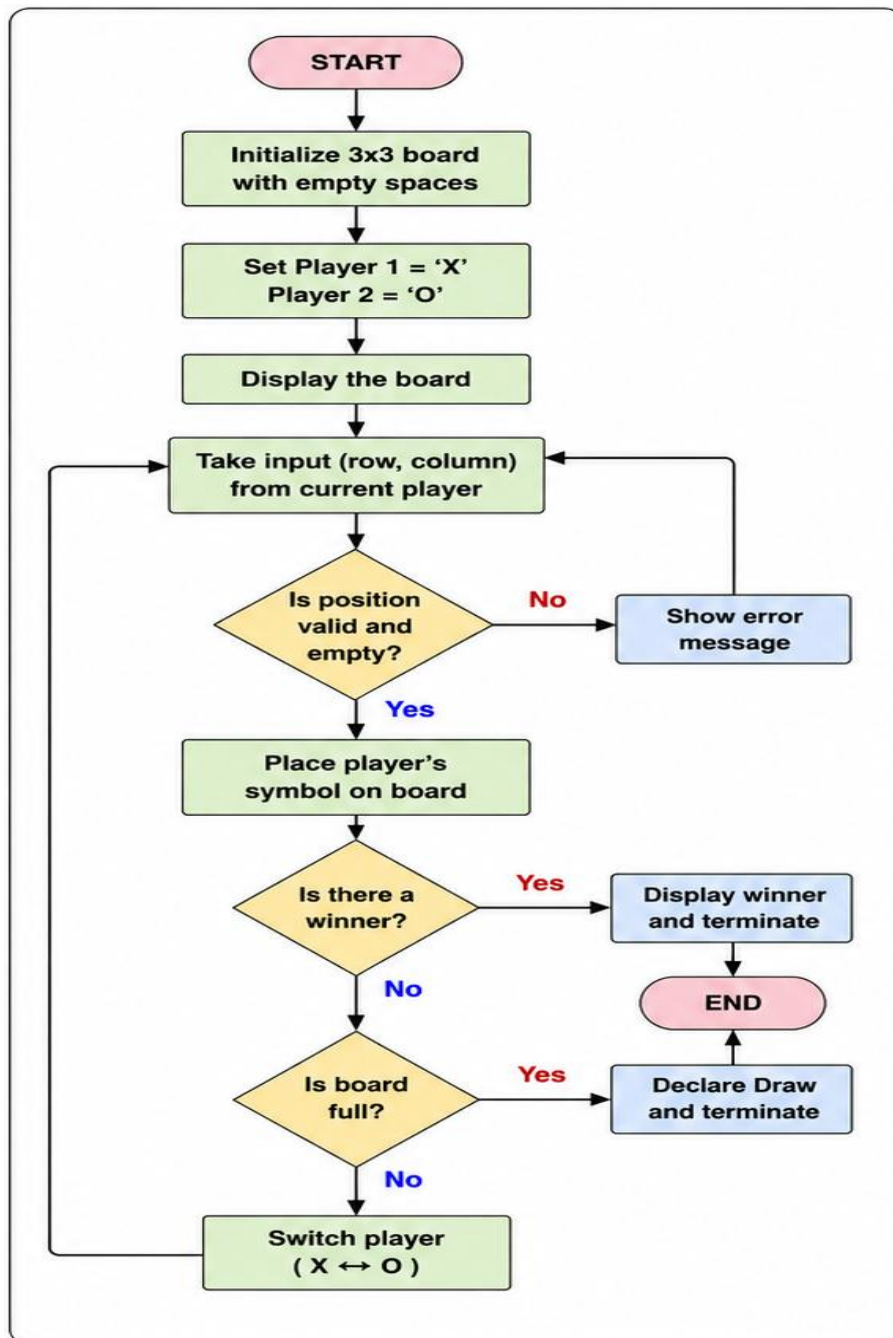
Objective :

- To develop a two-player Tic Tac Toe game.
- To implement game logic using Python.
- To validate player moves.
- To check winning and draw conditions.
- To switch turns between the two players.

Algorithm:

1. Initialize a 3×3 board with empty spaces.
2. Assign **X** to Player 1 and **O** to Player 2.
3. Display the current board.
4. Get the row and column from the current player.
5. Check whether the entered position is valid and empty.
6. Place the player's symbol on the board.
7. Check all rows, columns, and diagonals for a winning condition.
8. If a player wins, display the winner and terminate.
9. If the board is full and there is no winner, declare a draw.
10. Otherwise, switch the player and continue the game.

Flowchart :



Code:

```
def display_board(board):  
    print("\n")  
    for row in board:  
        print(" | ".join(row))
```

```

    print("-" * 9)
def check_winner(board, player):
    for row in board:
        if all(cell == player for cell in row):
            return True
    for col in range(3):
        if all(board[row][col] == player for row in range(3)):
            return True
    if all(board[i][i] == player for i in range(3)):
        return True
    if all(board[i][2 - i] == player for i in range(3)):
        return True
    return False
def board_full(board):
    for row in board:
        if " " in row:
            return False
    return True
def tic_tac_toe():
    board = [[" " for _ in range(3)] for _ in range(3)]
    players = ["X", "O"]
    current_player = 0
    print("TIC TAC TOE GAME")
    while True:
        display_board(board)
        player = players[current_player]

```

```

print("Player", player, "turn")
try:
    row = int(input("Enter row (0-2): "))
    col = int(input("Enter column (0-2): "))
except ValueError:
    print("Invalid input! Enter numbers only.")
    continue
if row not in range(3) or col not in range(3):
    print("Invalid position! Enter values from 0 to 2.")
    continue
if board[row][col] != " ":
    print("Position already occupied! Try again.")
    continue
board[row][col] = player
if check_winner(board, player):
    display_board(board)
    print("Player", player, "wins!")
    break
if board_full(board):
    display_board(board)
    print("Game Draw!")
    break
current_player = 1 - current_player
tic_tac_toe()

```

Output:

Output

TIC TAC TOE GAME

```
| | |
-----
```

```
| | |
-----
```

```
| | |
-----
```

Player X turn

Enter row (0-2): 0

Enter column (0-2): 0

Output

```
X | | 
-----
```

```
| | |
-----
```

```
| | |
-----
```

Player O turn

Enter row (0-2): 1

Enter column (0-2): 0

```
X | | 
-----
```

```
O | | 
-----
```

```
| | |
-----
```

Player X turn

Output

```
Enter row (0-2): 0
Enter column (0-2): 1
```

```
X | X |
-----
O |   |
-----
  |   |
-----
```

```
Player O turn
Enter row (0-2): 1
Enter column (0-2): 1
```

Output

```
X | X |
-----
O | O |
-----
  |   |
-----
```

```
Player X turn
Enter row (0-2): 0
Enter column (0-2): 2
```

```
X | X | X
-----
O | O |
-----
  |   |
-----
```

```
Player X wins!
```

Result :

The Tic Tac Toe game was successfully implemented in Python. The program logically validates player moves, alternates between players, checks rows, columns and diagonals for a winner, and identifies a draw when the board is full.